

NATIONAL INSTITUTE OF TECHNOLOGY, ANDHRA PRADESH

Department of Electronics and Communication Engineering

ECM-402: AI for Signal Processing (Ist Semester 2026-27)

Lab 1 (Wireless Channel Modeling using Linear Regression)

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- **Objective:** Implement a MyLinearRegression class (NumPy only) and use it to estimate a log-distance path-loss model from given measurements.
- **Theory:** Consider the following scenario
 - Scenario: Urban cellular environment at 3.5 GHz (n78 band)
 - Transmitter: 5G base station at height: 12m
 - Receiver: Mobile device at height: 1.5m
 - Reference distance: $d_0 = 1\text{m}$

The equation for the CI model is given as

$$\text{PL}^{\text{CI}}(d)[\text{dB}] = \text{FSPL}(d_0)[\text{dB}] + 10n\log_{10}(d) + \mathcal{X}_\sigma$$

where n denotes the single model parameter, the path loss exponent, with $10n$ describing rate of increase of path-loss. d is the distance between transmitter and receiver. $\text{FSPL}(d_0)$ denotes the free space path loss in dB at a T-R separation distance of d_0 . $\text{FSPL}(d_0)$ is given as

$$\text{FSPL}(d_0)[\text{dB}] = 20\log_{10}\left(\frac{4\pi d_0 f}{c}\right)$$

where c is the speed of light and f_c is the carrier frequency. $\mathcal{X}_\sigma$ is the zero-mean Normal random variable with standard deviation $\sigma_{\mathcal{X}}$ (in dB), representing large-scale shadowing.

You are provided with measurement data (pathloss.txt) containing distance (d) vs. measured path loss PL(dB).

- **Tasks:**

- (1) **Implement Linear Regression:** Write a class (MyLinearRegression) implementing (simple) linear regression algorithm. Provide three methods to estimate parameters for the linear regression model.
 - (a) Closed-form (Normal equation)
 - (b) Batch Gradient Descent (iterative)
 - (c) Stochastic Gradient Descent (SGD) (iterative)

Provide a method option for user to specify any of the three approaches.

- (2) **Estimate FSPL and Path-loss Exponent Jointly:** Choose a reference distance d_0 (e.g., $d_0 = 1$ m). Rearranged the linear model as

$$y = \underbrace{FSPL(d_0)}_{\text{intercept}} + \underbrace{10n}_{\text{slope}} \cdot \log_{10}(d/d_0) + \varepsilon.$$

Compute the model parameters $FSPL(d_0)[\text{dB}], n$ using the linear regression class implemented in (1). Chose the appropriate learning rate and number of epochs for iterative methods. Sketch the training loss vs epochs plot for both the iterative approaches and comment on the results.

- (3) **Estimate Shadowing Variance :** Compute residuals: $\varepsilon_i = y_i - \hat{y}_i$. Estimate the shadowing variance as

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N \varepsilon_i^2$$

Provide a table of estimated parameters $(FSPL(d_0), n, \sigma)$.

- (4) **Visualization :** Plot the measured path loss vs $\log_{10}(d)$. Add fitted models using all three approaches on the same graph. Comment on which model fits the measured data better.
- (5) **Use the Learned Model :** Using the parameters obtained in (2) to compute the received power (in dBm) at 2Km if the transmit power is 100 watt. What is the probability that the received power at $d=2\text{KM}$ will be larger than -95 dBm.

Submission Guidelines : You can solve problems either in MATLAB or PYTHON.

MATLAB Submission: Write a MATLAB script Lab1.m and all the function files (if any) required in the solution. Put all the files in a folder. Lab1.m file should have proper comments explaining the problem statement and solutions (code) for each problem. Use Publish button with PDF option to create a PDF of your solution. Zip everything as your Rollno.zip and submit.

See https://in.mathworks.com/help/matlab/matlab_prog/publishing-matlab-code.html

Python Submission: Write a Jupyter notebook Lab1.ipynb containing solution of all the problems in the sheet. Add problem statement/extra text for each problem to Jupyter notebooks using Markdown cells. Convert the notebook into a pdf document using notebook-as-pdf package. Zip your notebook,pdf and result figures as your Rollno.zip and submit.

See <https://builtin.com/articles/jupyter-notebook-to-pdf>.